SECD2613 SYSTEM ANALYSIS AND DESIGN

SESSION 2025/2026 – SEMESTER 2

PHASE 1: PROJECT PROPOSAL AND PLANNING

FlexHub AI Marketplace [FlexServe]

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1.0 Introduction

Our team consists of five Bioinformatics students from Universiti Teknologi Malaysia (UTM). We are creating our project as part of the System Analysis and Design subject. Our project idea is named FlexServe. It is a platform aimed at assisting professional service providers, such as freelance make-up artists, private tutors, and others.

Many freelance professionals use informal solutions, which may lead to problems with scheduling, as well as communication and booking issues. For customers, it may be difficult to find the suitable freelancer since they do not have access to one platform to get the required information.

To solve all these problems, we suggest using our platform FlexServe. First of all, it will simplify the search for freelancers. Also, there is an opportunity to communicate with the selected freelancer and book the service. The main idea of FlexServe is that the system will be able to match the customer with the freelancer based on the type of needed services and their free time.

Thus, FlexServe is a platform to solve all possible problems in terms of efficient finding of freelance professionals for a service. We focused on those service providers who offer an hourly or session basis service provision.

2.0 Background Study

In the last few years, service providers have turned to social media sites to market their services. Sites like Instagram and WhatsApp have been used to display portfolios, reach out to customers, and market available services. There is a large variety of services offered by professional service providers. Makeup services, tutoring, consulting, event management, photography, and technical support freelance services are among others that professionals offer. The services are done using different skills, schedules, prices, instruments, and approaches depending on customers' needs. This means that flexibility is an important aspect of the service sector for freelancers or independent service providers to operate effectively.

The increased reliance on technology has changed expectations of customers. Customers today expect quick response time, effective booking system, competitive pricing, and convenience while accessing professional services online. Most people appreciate a system that allows them to compare various service providers, review portfolios, and book services without contacting several individuals separately. However, many freelance service providers still use the booking procedures such as the direct messaging apps or social media messaging services. This causes many issues like schedule management problems, delays in responses, double bookings, and poor communication between customers and service providers. Customers also face challenges in finding good service providers due to fragmented information across various channels.

Additionally, with increasing demand for flexible services, the need for an automated and efficient booking system becomes crucial. Since there is a rise of flexible hourly and per-session-based services, having an efficient system for booking could help increase efficiency during the booking process. Hence, to overcome such concerns and make it easier for people to get access to good service providers, FlexServe system is being designed to be an intelligent and integrated platform that will be helpful for managing professional services with flexible schedule management and smart booking management.

3.0 Problem Statement

One of the biggest challenges that are being encountered in the professional services industry is the lack of a single platform where the customer can browse and book services from various professionals. Customers have to go through several social media sites and messaging applications just to be able to hire appropriate service providers, making the entire process inefficient and cumbersome since data needs to be collected from various sources.

The process involved is mostly manual where customers need to communicate with service providers via phone calls and chats to enquire about pricing, availability, services, and other relevant details. This manual method would cause delay of services since there high possibility of double bookings and misunderstandings between both parties. Moreover, some customers experience issues when trying to compare different service providers due to the lack of a proper mechanism to check their portfolios, ratings, prices, and availability.. The operators of the services can also face difficulties regarding their scheduling, client registration, and appointment timing. As a result of not using any scheduling systems, they may mistakenly make overlapping appointments or not leave enough space between them, thus resulting in poor service quality and customer experience.

Additionally, there is no intelligent recommender system used within the existing booking systems. Clients usually waste a lot of time looking for appropriate service providers without getting any recommendations based on their preferences, budget, location, and booking record. In order to resolve the problems mentioned above, FlexServe is suggested as a solution to become an AI-enabled service management system that will facilitate booking process and will include smart scheduling, availability check, and AI recommendations.

4.0 Proposed Solutions

4.1 Operational Feasibility

Operational feasibility for FlexServe is based on the ability of the project team in coordinating efforts to make sure that the software works seamlessly and meets the requirements of the users. Each team member will play a crucial role in creating the platform efficiently and feasible through the implementation process and during using the software.

The Project Manager will be involved in coordinating the activities of the project and monitoring the progress of development of the system. The Operation Specialist will ensure coordination and management of booking activities, scheduling activities, and user interaction in a consistent manner to ensure efficient operation of the platform. The Technical Analyst is required to analyze the technical aspects of the system, its feasibility, and help in the implementation of an appropriate solution from the technical aspect. On the other hand, the Financial Analyst will analyze the feasibility of the system based on a financial point of view.

FlexServe is also utilized to ease management and greater efficiency for both customers and service providers, since it integrates all processes related to booking and scheduling into a single interface. This system has AI-based recommendation features, which will help in finding appropriate service providers for the clients according to their needs and preferences.

Finally, the software needs to have a user-friendly interface, to cater to users with little knowledge about computers to use the software seamlessly.

4.2 Technical Feasibility

Technical feasibility for the creation of FlexServe can be done using current technology, development environment, and cloud technology that are fit for academic system development. FlexServe can be created using web and mobile technology that can utilise databases and AI integration. The role of the Technical Analyst is to conduct an analysis on the system requirements, assess the feasibility of the proposed solution, and assist in the selection of appropriate technologies such as frameworks, databases, and AI APIs. Current AI APIs and cloud technologies can be used to streamline the development process. Some of the key aspects included in FlexServe are booking management, scheduling, availability management, payment management, and intelligent matching of clients with service providers.

1. Scalability

   Through cloud-based microservices, FlexServe will be able to accommodate more numbers of users and their service requests effectively. In the event that there is any peak load during holidays, examination times, or wedding occasions, the booking and recommendation services of FlexServe will still operate efficiently.
2. Data Privacy

   Technological security controls are essential since FlexServe holds confidential user information such as personal information, transaction history, chat logs, and financial transactions. It is important to have encryption and authentication mechanisms to safeguard this information.
3. Performance

   It is important that such features should be implemented efficiently with rapid response times so that users are not disappointed with the platform performance. It is possible with today's technological advancements.

4.3 Economic Feasibility

Economic feasibility for FlexServe involves assessing whether the total cost of developing and implementing the system outweighs its benefits. The project will be economically feasible since the system can cut down the manual work involved in managing and booking services.

4.4 Cost-Benefit Analysis (CBA)

Estimated Cost	
Hardware	RM40 000
Software	RM 6 500
Consultant	RM18 000
Training	RM14 000
Supplies	RM 2 400 per year
IS Support	RM10 000 per year
Maintenance	RM 2 500 per year

Estimated Benefits	
Transaction Commision	RM58 500 per year
Efficiency Gains	RM 5 000 per year
Premium Listing Fees	RM 1 800 per year

Assumptions	
Discount Rate	10%

Assumptions	
Sensitivity Factor (cost)	1.1
Sensitivity Factor (benefits)	0.9
Annual change in production costs	7%
Annual change in benefits	10%

Cost-Benefit Analysis (CBA) - Costs

Cost	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Development Costs ● Hardware ● Software ● Consultant ● Training	 44 000 7 150 19 800 15 400					
Total	86 350					
Production Costs ● Supplies ● Network Personnel ● Maintenance		 2 640 11 000 2 750	 2 825 11 770 2 943	 3 023 12 594 3 149	 3 235 13 476 3 369	 3 461 14 419 3 605
Annual Production Costs Present Value (PV)		16 390 14 900	17 538 14 494	18 766 14 099	20 080 13 715	21 485 13 340
Accumulated Costs		101 250	115 744	129 843	143 558	156 898

Cost-Benefit Analysis (CBA) - Benefits, Gain/Loss and Profitability Index

Benefits	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Transaction Commision Efficiency Gains Premium Listing Fees		52 650 4 500 1 620	57 915 4 725 1 701	63 707 4 961 1 786	70 078 5 209 1 875	77086 5 469 1 969
Total Present Value (PV)		58 770 53 427	64 341 53 174	70454 52 933	77 162 52 703	84524 52 483
Accumulated Benefits		53 427	106 601	159 534	212237	264 720
Gain/Loss		(47 823)	(9 143)	29 691	68 679	107 822
Profitability Index	1.25					

Profitability index = 1.25, showing that it is a good investment because the profitability index is more than one.

The CBA for FlexServe demonstrates that the company has a very good financial outlook, with a break-even point in the third year of operation. Although the total amount required for the initial cost of development, which is RM 86,350, is quite high, involving advanced hardware and AI technology, as well as labor costs, it is important to invest in order to guarantee scalability and data protection. The steady rise in benefits over the first five years demonstrates expected user engagement and subscriptions by the service providers. The positive Net Present Value (NPV) and Return on Investment (ROI) indicate that the project is economically feasible.

5.0 Objectives

1. As a centralized service to allow new and existing service providers to highlight their skill sets, areas of expertise, and services offered by them in several fields of professional work.

2. As a management service for both the experienced and novice service providers, to enable them to easily promote and organize themselves in a more systematic way.

3. As a means of matching the users' preferences with appropriate service providers using AI technology.

4. To allow the users to gain access to the profile of service providers, which includes their qualifications, price, portfolio, and availability.

5. To facilitate better interaction between the users and service providers in making bookings and minimizing message overlaps.

6. To make bookings easier for the users by allowing flexible booking periods like hour-wise or per session booking periods.

6.0 Scope of the Project

FlexServe is a proposed web-based application platform for facilitating booking and discovery of service providers. In particular, the purpose of the FlexServe is to simplify booking a professional service and to ensure that all parties involved have an experience. FlexServe facilitates an easy service discovery by consolidating all relevant information and offering suggestions.

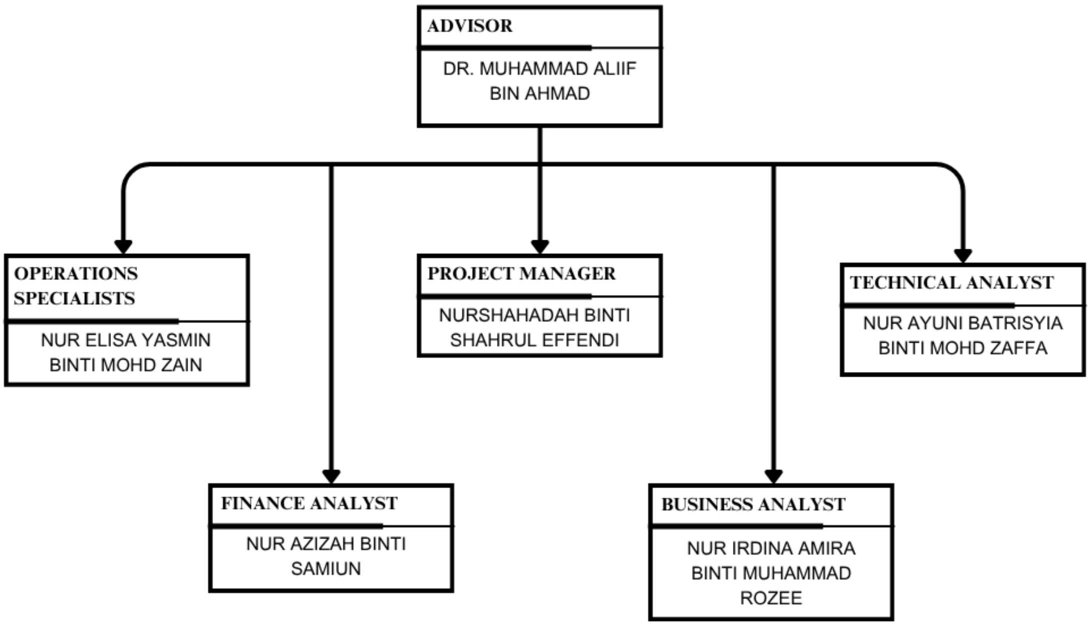
To gain access to FlexServe users need to register themselves providing the following details: name, email address, phone number and password. There are two main types of users in FlexServe, the Clients and Service Providers. Both types have their unique way of interacting with the FlexServe and have access to different aspects of it.

The clients can use FlexServe to find various professional services such as tutoring, consulting, makeup artistry among others. In doing so, the clients will be able to learn more about various professionals, the range of services offered, the prices charged and client testimonials regarding their work. Further, clients will be able to apply filters in order to locate a suitable professional within the specified budget. Having identified an interesting person, the user will be notified on whether that person is available and will be able to book him/her and pay for the service.

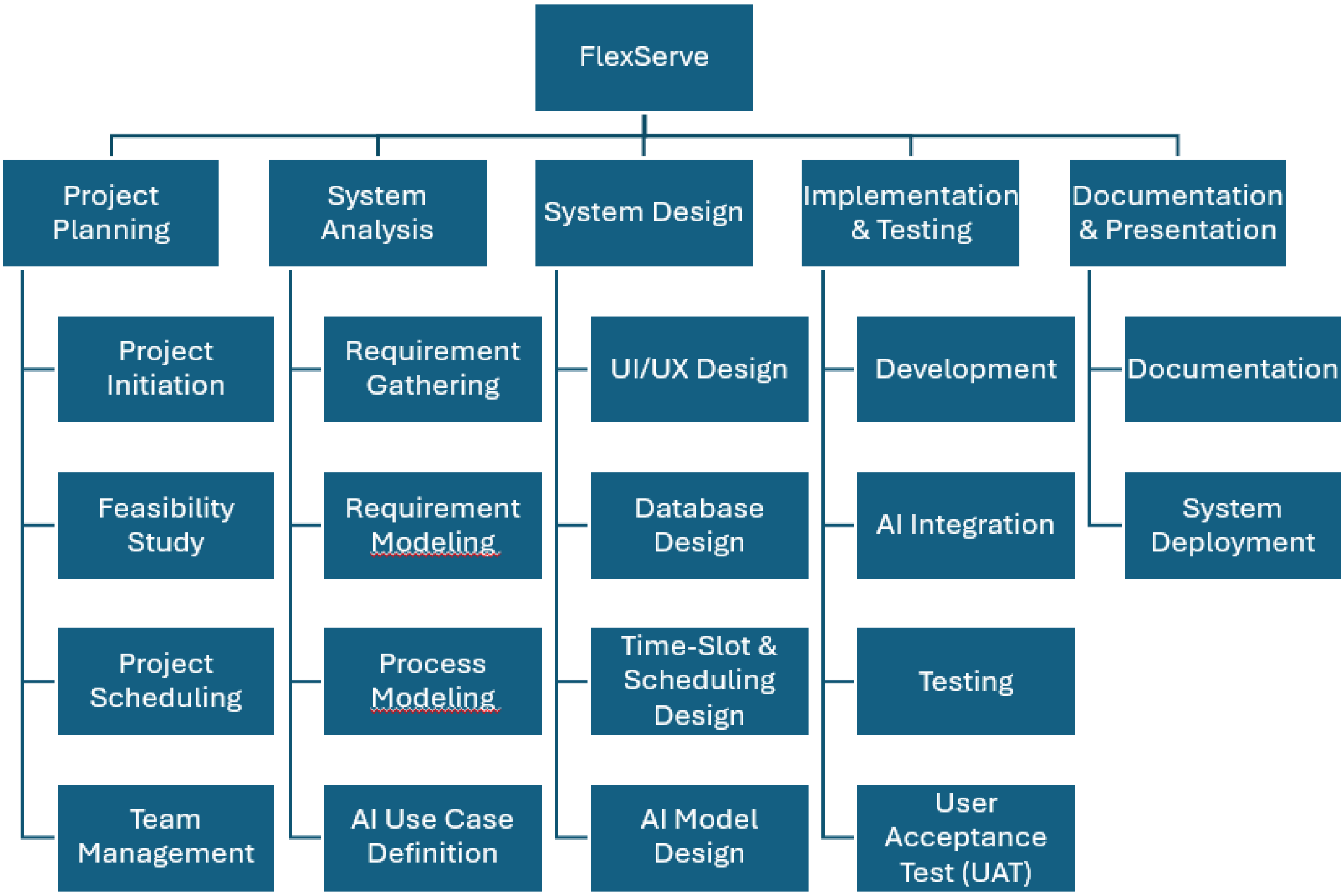
FlexServe allows service providers to manage their operations. They can modify their details, add their services, and adjust prices. They can also be notified whenever a client wishes to hire them, and they can control their calendar by accepting or rejecting booking requests. Freelancers will easily keep track of their appointments and ensure that clients do not experience any difficulties when seeking services from them. FlexServe is an application that facilitates ease and convenience for both service providers and clients. FlexServe is a system that simplifies the whole process of service bookings and hiring. FlexServe makes the life of clients and service providers much simpler.

7.0 Project Planning

7.1 Human Resource

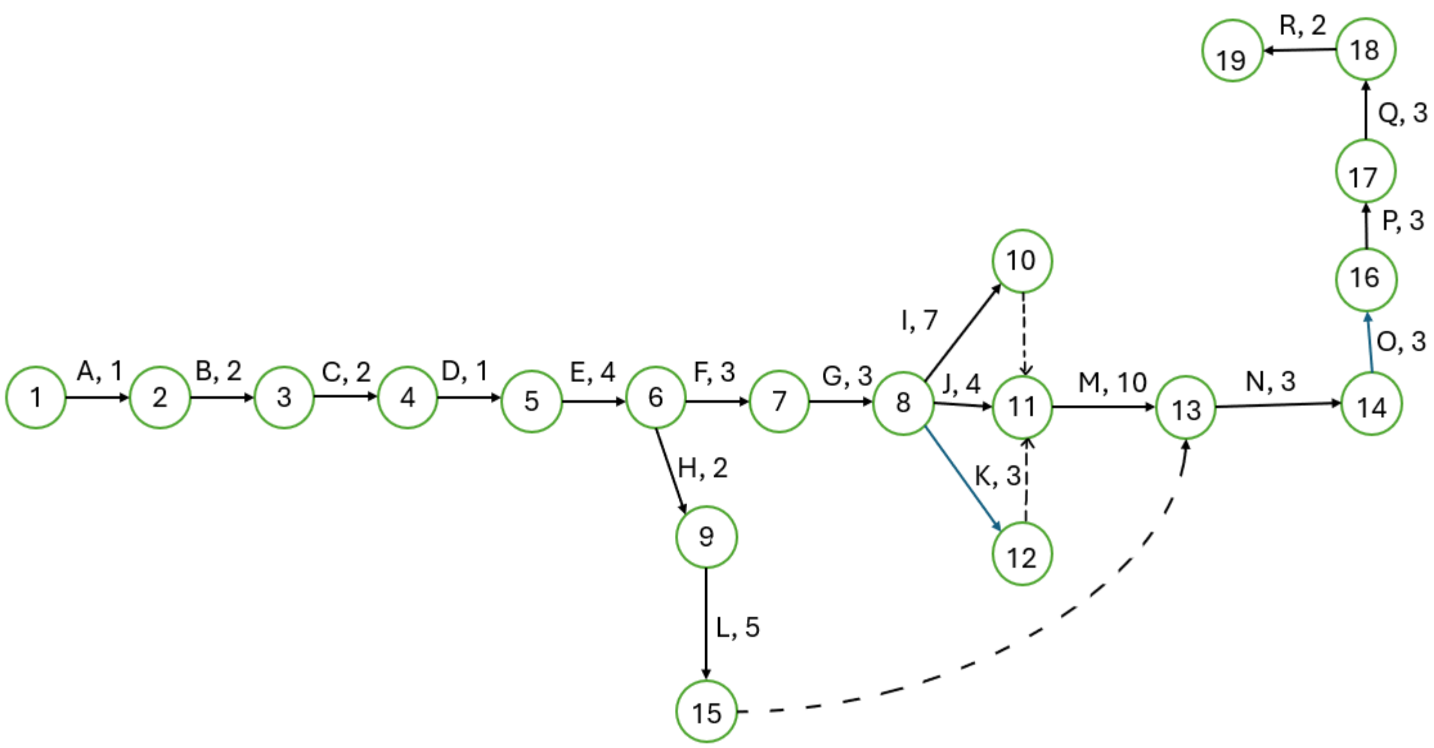


7.2 Work Breakdown Structure



7.3 PERT Chart

Description	Task	Must Follow	Time (Days)
Project Initiation	A	None	1
Feasibility Study	B	A	2
Project Scheduling	C	B	2
Team Management	D	C	1
Requirements Gathering	E	D	4
Requirements Modeling	F	E	3
Process Modelling	G	F	3
AI Use Case Definition	H	E	2
UI/UX Design	I	G	7
Database Design	J	G	4
Time-Slot & Scheduling Design	K	G	3
AI Model Design	L	H	5
Development	M	I, J, K	10
AI Integration	N	L, M	3
Testing	O	N	3
UAT	P	O	3
Documentation	Q	P	3
System Deployment	R	Q	2



Path 1: A-B-C-D-E-F-G-I-M-N-O-P-Q-R

Length 1: 1+2+2+1+4+3+7+10+3+3+3+3+2 = 47 Days

Path 2: A-B-C-D-E-F-G-J-M-N-O-P-Q-R

Length 2: 1+2+2+1+4+3+4+10+3+3+3+3+2 = 44 Days

Path 3: A-B-C-D-E-F-G-K-M-N-O-P-Q-R

Length 3: 1+2+2+1+4+3+3+10+3+3+3+3+2 = 43 Days

Path 4: A-B-C-D-E-H-L-N-O-P-Q-R

Length 4: 1+2+2+1+4+2+5+3+3+3+3+2 = 31 Days

Since critical path are the longest path for PERT diagram, Path 1 are the longest path, which will take 47 days to complete.

Critical Path: Path 1 (47 Days)

7.4 Gantt Chart

Activity	Days started	Duration	Days Ended
Project Initiation	1	1	1
Feasibility Study	2	2	3
Project Scheduling	4	2	5
Team Management	6	1	6
Requirements Gathering	7	4	10
Requirements Modeling	11	3	13
Process Modelling	14	3	16
AI Use Case Definition	11	2	12
UI/UX Design	17	7	23
Database Design	17	4	20
Time-Slot & Scheduling Des	21	3	23
AI Model Design	13	5	17
Development	24	10	33
AI Integration	34	3	36
Testing	37	3	39
UAT	40	3	42
Documentation	43	3	45
System Deployment	46	2	47

The Gantt chart for the FlexServe project includes the project schedule with task duration, dependencies, and deadlines. The chart assists the team in monitoring project progress to ensure timely completion of the project.

8.0 Benefit and Overall Summary of Proposed System

The FlexServe system is a tool that helps people who work for themselves like freelancers and their clients. It makes it easier for them to find each other and work together. The FlexServe system has a way of matching people with the right expert for the job, which saves a lot of time and effort.

One of the problems that FlexServe solves is when people have schedules that do not work well together. The FlexServe system is like a place where clients can go to find the help they need instead of looking all over the internet. This way clients do not have to waste a lot of time looking at lots of social media posts that are not organized.

The FlexServe system also helps the people who are providing the services. They do not have to have a lot of social media accounts to try to find customers. The FlexServe system helps them get their name there and find people who need their help. It also helps them keep track of their schedules, which can be really hard when they have to travel to places.

The FlexServe system makes it easy for professionals and their clients to talk to each other. If someone is not available to help, the FlexServe system will suggest someone who can. This way people do not have to send a lot of messages and they can always find someone to help them.

So the FlexServe system is a good solution for the problems that people have when they are trying to find help or provide services. It makes it easy for freelancers to focus on their work. It makes it easy for clients to find the help they need. The FlexServe system is like a one stop shop for people who need help and for people who provide services. The FlexServe system helps everyone work together smoothly and it makes the FlexServe system a great tool, for the users.

9.0 GitHub repository and Projects

URL of the GitHub Repository:

https://github.com/nurshahadahshahruleffendi/Error_404_Project1_SAD_20252026

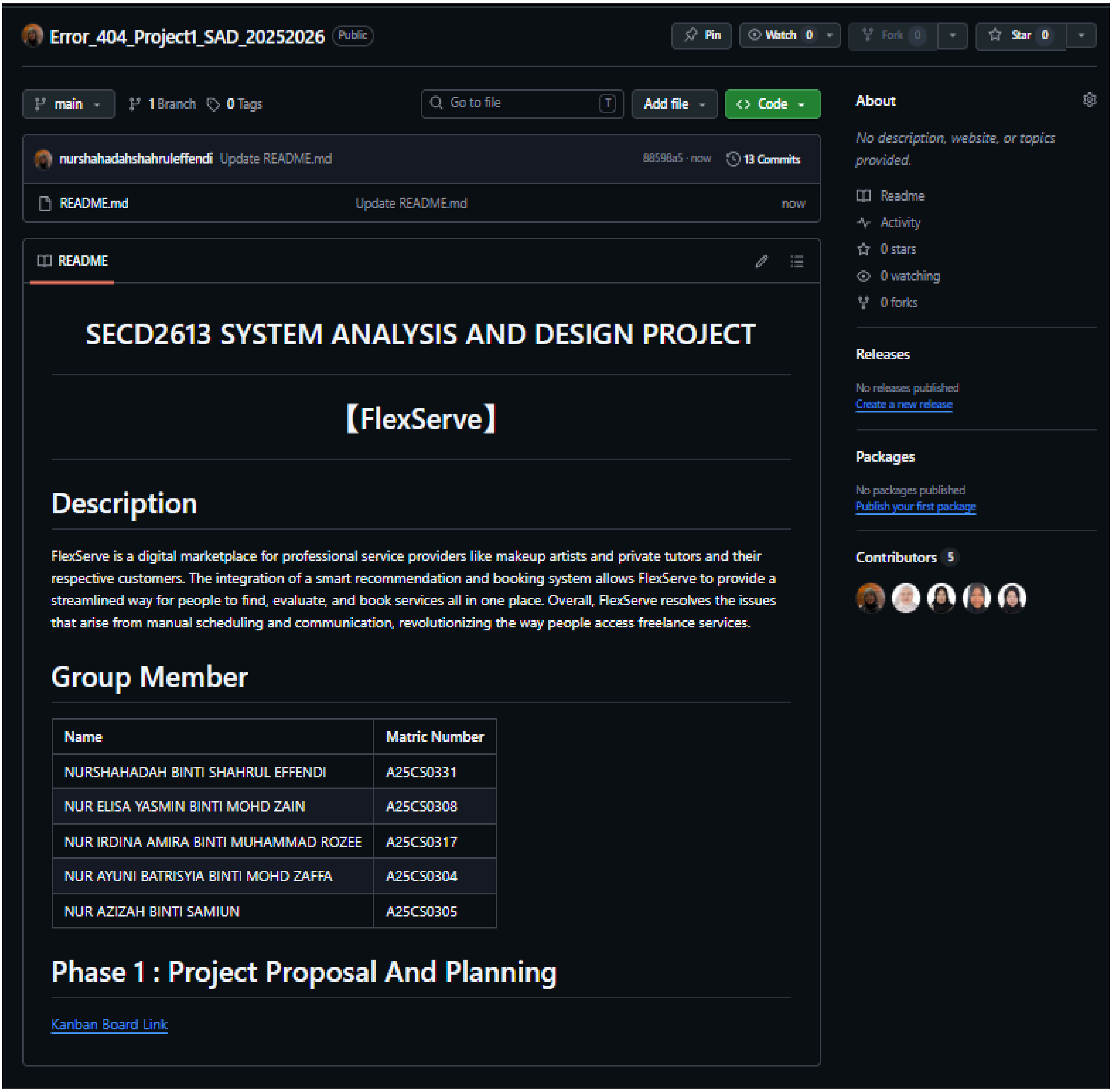


Figure 1: Respiratory Snapshots

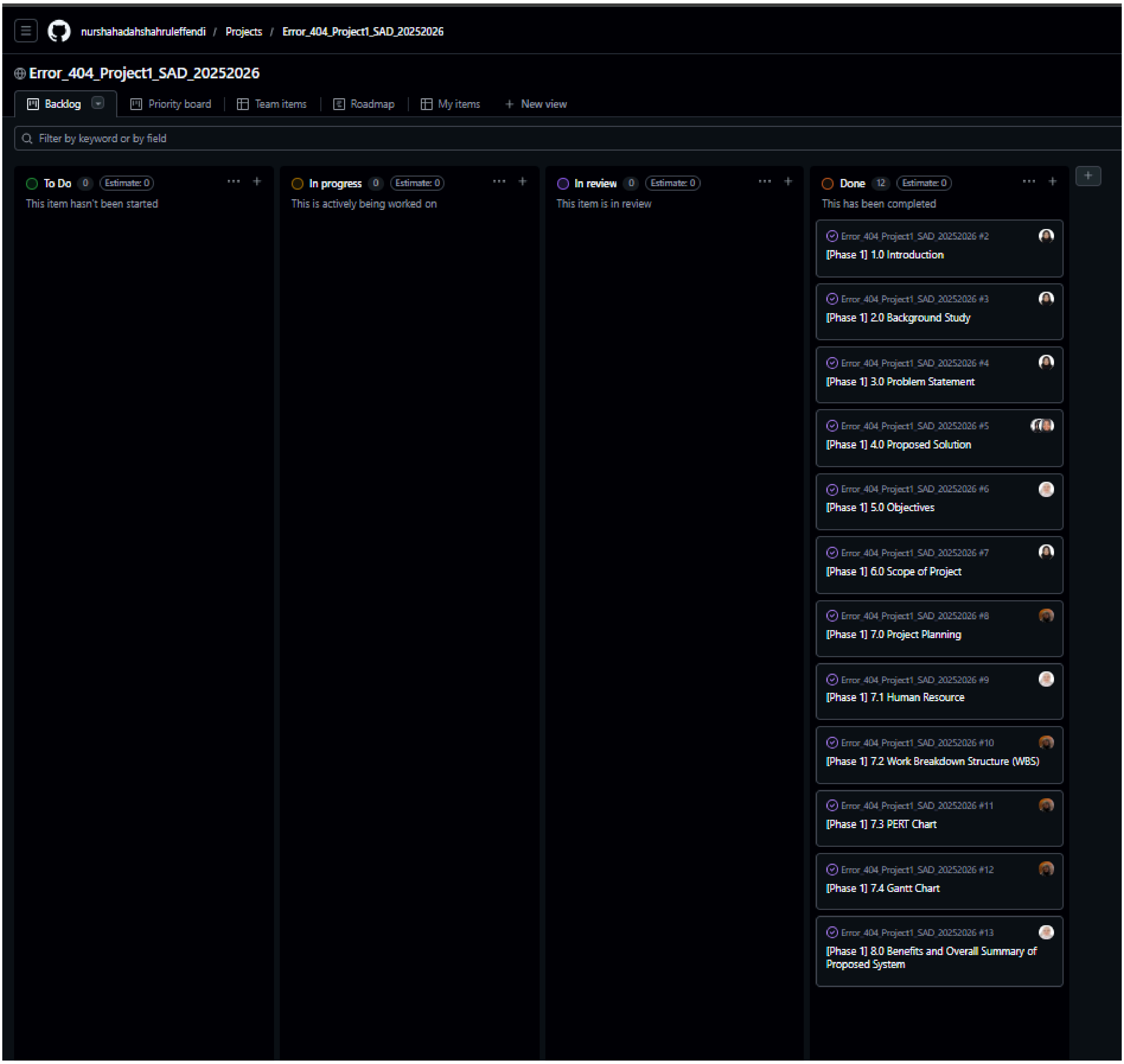


Figure 2: Kanban Board Integration

Version Control Practices:

We use the GitHub Kanban board to visualize the workflow of our project. This can make sure that every task is tracked from “To-Do” to “Done” preventing restriction in the development cycle. Team members can check the project's status at any time without needing constant meetings. This visibility helps us identify if anyone is stuck or if a task is taking longer than planned. We use the GitHub Kanban board to divide tasks among team members. Each card is assigned to a specific person, ensuring that everyone is clear about their responsibilities and tasks needed to be done. This prevents overlapping work and to make sure each member takes accountability across the team. The GitHub repository serves as the single source for all of our project documentation. The “Readme” file and Kanban Board is regularly updated to reflect the current project status, team roles, and repository structure for full transparency.